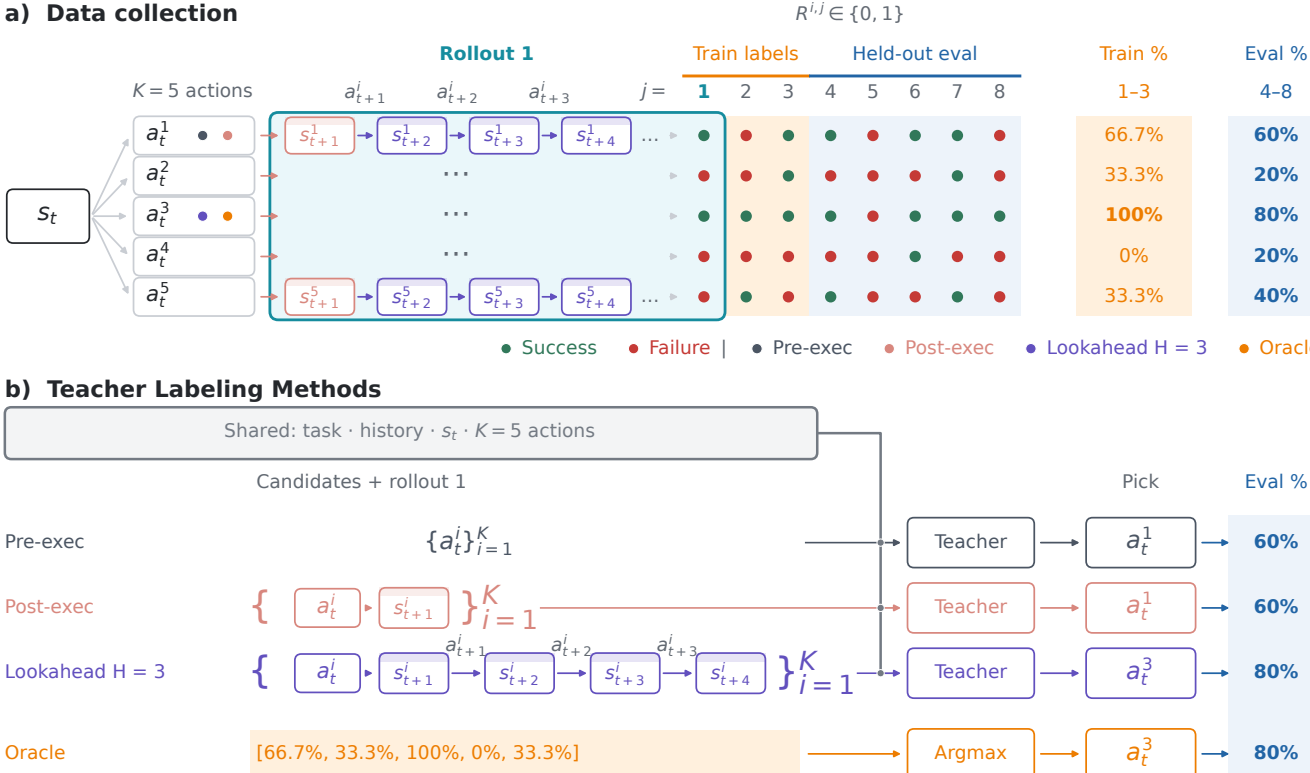
a) Data collection
$R^{i,j} \in \{0, 1\}$
Rollout 1
Train labels
Held-out eval
Train %
Eval %
$K = 5$ actions
a^i_{t+1} a^i_{t+2} a^i_{t+3}
$j =$ 1 2 3 4 5 6 7 8
s_t
a^1_t a^2_t a^3_t a^4_t a^5_t
s^1_{t+1} s^1_{t+2} s^1_{t+3} s^1_{t+4} …
s^5_{t+1} s^5_{t+2} s^5_{t+3} s^5_{t+4} …
1–3
4–8
66.7% 60%
33.3% 20%
100% 80%
0% 20%
33.3% 40%
Success Failure | Pre-exec Post-exec Lookahead H = 3 Oracle
b) Teacher Labeling Methods
Shared: task · history · s_t · $K = 5$ actions
Candidates + rollout 1
Pick
Eval %
Pre-exec
$\{a^i_t\}^K_{i=1}$
Post-exec
$\{ a^i_t \to s^i_{t+1} \}^K_{i=1}$
Lookahead H = 3
$\{ a^i_t \to s^i_{t+1} \to s^i_{t+2} \to s^i_{t+3} \to s^i_{t+4} \}^K_{i=1}$
Oracle
[66.7%, 33.3%, 100%, 0%, 33.3%]
Teacher
Teacher
Teacher
Argmax
a^1_t 60%
a^1_t 60%
a^3_t 80%
a^3_t 80%